MATH 6660 — Homework 4

2D Cartesian FEM, Problem 4(a)–(c)

Fall 2026

Relation to the lectures. Lecture 13, Part III, does exactly this problem with $u = 0$ on $\partial\Omega$: product rule, divergence theorem, $V = H_0^1$, and the four Lax–Milgram checks with a Poincaré constant left unnamed. Two things change here. The boundary value is $g \neq 0$, which is Lecture 5's nonhomogeneous Dirichlet case (trial set $\neq$ test space, lifting). And the elements are squares with the bilinear interpolant the problem gives; their basis turns out to be products of Lecture 4's 1D hats. The Poincaré constant for the square is computed by Lecture 8's 1D argument.

Problem 4(a)

Problem. Recast this problem as a variational problem. Clearly state the test and trial function spaces.

The strong problem is

$$-\nabla \cdot \nabla u = -u_{,xx} - u_{,yy} = f(x,y) \quad \text{in } \Omega = (-1,1)\times(-1,1), \qquad u = g \quad \text{on } \partial\Omega. \tag{1}$$

Step 1 — multiply by v, integrate, move one derivative [Lecture 13, Part III: product rule, then divergence theorem]

- Multiply (1) by a test function v and integrate over Ω:

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{2}$$

- Product rule, $\nabla\cdot(v\nabla u) = \nabla v\cdot\nabla u + v\,\Delta u$, so the left side of (2) is

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y. \tag{3}$$

- Divergence theorem on the last term, $\boldsymbol{n}$ the outward unit normal: $\iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial\Omega} v\,\nabla u\cdot\boldsymbol{n}\,\mathrm{d}s$. Put this into (3) and (3) into (2):

$$\iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y \;-\; \oint_{\partial\Omega} v\,\underbrace{\nabla u\cdot\boldsymbol{n}}_{\text{normal flux, not given}}\,\mathrm{d}s \;=\; \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{4}$$

Step 2 — kill the boundary term [Lecture 5: Dirichlet = essential, the test function vanishes there]

- On all four sides the data is the *value* $u = g$. The flux $\nabla u\cdot\boldsymbol{n}$ there is not known — it is the reaction to holding u at g. So the boundary integral in (4) cannot be evaluated.
- The only way to remove it is to demand $v = 0$ on $\partial\Omega$. Then (4) becomes

$$\boxed{\iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y} \tag{5}$$

Step 3 — the trial set and the test space [Lecture 5: $u(0) = \alpha \neq 0$ gives trial $V \neq$ test V_0; Lecture 13, eq. (1.254), for the 2D integrals]

- (5) needs $\iint_\Omega |\nabla u|^2$ and $\iint_\Omega |\nabla v|^2$ finite, so both spaces are built on $\iint_\Omega \left[v^2 + |\nabla v|^2\right] \mathrm{d}x\,\mathrm{d}y < \infty$, as in Lecture 13.
- The unknown u must also satisfy the essential condition $u = g$; the test function must satisfy $v = 0$ (Step 2):

$$\boxed{\begin{aligned} \text{trial:}\quad & V_g = \left\{ v : \iint_\Omega \left[v^2 + |\nabla v|^2\right] \mathrm{d}x\,\mathrm{d}y < \infty, \quad v = g \text{ on } \partial\Omega \right\}, \\ \text{test:}\quad & V_0 = \left\{ v : \iint_\Omega \left[v^2 + |\nabla v|^2\right] \mathrm{d}x\,\mathrm{d}y < \infty, \quad v = 0 \text{ on } \partial\Omega \right\} = H_0^1(\Omega), \end{aligned}} \tag{6}$$

  with $|\nabla v|^2 = (v_{,x})^2 + (v_{,y})^2$.
- They differ because $g \neq 0$ (in part (d), $g = 1$). Exactly Lecture 5's warning: V_g is not a vector space — $u_1 + u_2$ has boundary value $2g$ — while V_0 is.

(Variational problem) Find $u \in V_g$ such that

$$\iint_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y \qquad \text{for all } v \in V_0,$$

with trial set V_g and test space $V_0 = H_0^1(\Omega)$ from (6).

Mechanics reading. A membrane with unit tension under pressure f, its rim held at height g. V_g: deflected shapes that honour the rim. V_0: virtual deflections, which may not move the rim. (5) is virtual work; the rim reaction does no virtual work because $v = 0$ there.

Problem 4(b)

Problem. Show that the variational problem has a unique solution by showing that it meets all of the criteria of the Lax–Milgram theorem.

Step 1 — put the problem in Lax–Milgram's form [Lecture 13: u and v in the same Hilbert space; Lecture 5: lift the boundary data]

- Lax–Milgram (Lecture 13) asks for *one* Hilbert space V with $u, v \in V$. In 4(a), $u \in V_g$ but $v \in V_0$. Lecture 5's fix: split off a known function that carries the boundary data.
- Pick any $G \in V_g$. (If V_g were empty there would be no candidate u at all.) For part (d), $g = 1$ and $G \equiv 1$ works: $\iint_\Omega [1^2 + 0]\,\mathrm{d}x\,\mathrm{d}y = 4 < \infty$.
- Write $u = w + G$. Then $w = g - g = 0$ on $\partial\Omega$, so $w \in V_0$. Substitute in (5) and move the known part to the right:

$$\underbrace{\iint_\Omega \nabla w \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{B(w,v)} = \underbrace{\iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla G \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{\widetilde{L}(v)} \qquad \forall\, v \in V := V_0 = H_0^1(\Omega), \tag{7}$$

  with the norm of Lecture 13, $\|v\|_V^2 = \|v\|^2 + \|\nabla v\|^2$, where $\|\cdot\|$ is the $L^2(\Omega)$ norm.

Step 2 — (S) and (CB) [Lecture 13, Part III, checks 1–2; Lecture 6: Cauchy]

- **(S)** $\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$ pointwise, so $B(w, v) = B(v, w)$.
- **(CB)**

$$|B(w,v)| \le \underbrace{\iint_\Omega |\nabla w|\,|\nabla v|\,\mathrm{d}x\,\mathrm{d}y}_{|\boldsymbol{a}\cdot\boldsymbol{b}|\le|\boldsymbol{a}||\boldsymbol{b}|} \le \underbrace{\|\nabla w\|\,\|\nabla v\|}_{\text{Cauchy}} \le \underbrace{\|w\|_V \|v\|_V}_{\|\nabla v\|^2 \le \|v\|^2 + \|\nabla v\|^2}, \qquad \beta^* = 1. \tag{8}$$

Step 3 — Poincaré on the square [Lecture 8, eqs. (1.127)–(1.128): the same three lines, in x, for each fixed y]

- Take $v \in V_0$ and fix y. $v(-1, y) = 0$ on the left side, so

$$v(x,y) = \int_{-1}^{x} v_{,x}(s,y)\,\mathrm{d}s. \tag{9}$$

- Bound it by the whole line and apply Cauchy to $1 \cdot |v_{,x}|$ on an interval of length 2:

$$|v(x,y)| \le \int_{-1}^{1} |v_{,x}(s,y)|\,\mathrm{d}s \le \underbrace{\Big(\int_{-1}^{1} 1^2\,\mathrm{d}s\Big)^{1/2}}_{\sqrt{2}} \Big(\int_{-1}^{1} v_{,x}(s,y)^2\,\mathrm{d}s\Big)^{1/2}. \tag{10}$$

- Square (10). The right side no longer depends on x, so integrating over $x \in (-1, 1)$ multiplies it by 2; then integrate over $y \in (-1, 1)$:

$$\boxed{\|v\|^2 = \iint_\Omega v^2\,\mathrm{d}x\,\mathrm{d}y \le \underbrace{2 \cdot 2}_{x\text{-length} \times (10)} \iint_\Omega (v_{,x})^2\,\mathrm{d}x\,\mathrm{d}y \le 4\,\|\nabla v\|^2} \tag{11}$$

  This is Lecture 13's multi-dimensional Poincaré, $\|v\| \le C\|\nabla v\|$, with $C = 2$.

Step 4 — (VE) and the bound on $\widetilde{L}$ [Lecture 13, Part III, checks 3–4]

- **(VE)** Split $B(v,v)$ in half and use (11) on one half:

$$B(v,v) = \|\nabla v\|^2 = \tfrac{1}{2}\|\nabla v\|^2 + \underbrace{\tfrac{1}{2}\|\nabla v\|^2}_{\geq \frac{1}{8}\|v\|^2 \text{ by (11)}} \geq \tfrac{1}{8}\big(\|\nabla v\|^2 + \|v\|^2\big) = \tfrac{1}{8}\|v\|_V^2, \qquad \beta_* = \tfrac{1}{8}. \tag{12}$$

  Same as Lecture 13's $\beta_* = \frac{1}{2}\min(1/C^2, 1)$ with $C = 2$.

- $\widetilde{L}$ **bounded**, by Cauchy on each term of (7):

$$|\widetilde{L}(v)| \leq \|f\|\,\|v\| + \|\nabla G\|\,\|\nabla v\| \leq \big(\|f\| + \|\nabla G\|\big)\|v\|_V. \tag{13}$$

  This needs $\iint_\Omega f^2 < \infty$. In part (d), f is continuous on the closed square, $e^{\sin \pi x \sin \pi y} \leq e$ and the brace is at most $2+2+1+1 = 6$, so $|f| \leq 6\pi^2 e$; and $\nabla G = 0$.

condition	holds because	constant	equation
(S) symmetric	$\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$	—	Step 2
(CB) bounded	Cauchy	$\beta^* = 1$	(8)
(VE) V-elliptic	Poincaré, $C = 2$	$\beta_* = 1/8$	(12)
$\widetilde{L}$ bounded	Cauchy	$\|f\| + \|\nabla G\|$	(13)

Table 1: The four Lax–Milgram conditions for (7) on $V = H_0^1\big((-1,1)^2\big)$.

Step 5 — conclusion, and uniqueness of u itself

- Table 1 meets every hypothesis of Lax–Milgram, so (7) has a unique $w \in V_0$, with $\|w\|_V \leq \widetilde{L}/\beta_* = 8\big(\|f\| + \|\nabla G\|\big)$. Then $u = w + G \in V_g$ solves the variational problem of 4(a).
- u does not depend on which G was picked. If $u_1, u_2 \in V_g$ both solve it, $d = u_1 - u_2$ has boundary value $g - g = 0$, so $d \in V_0$, and subtracting the two problems gives $B(d,v) = 0$ for all $v \in V_0$. Take $v = d$ and use (12):

$$\tfrac{1}{8}\|d\|_V^2 \leq B(d,d) = 0 \qquad \Longrightarrow \qquad d = 0, \quad u_1 = u_2. \tag{14}$$

Conclusion. With $u = w + G$, the problem for w on $H_0^1(\Omega)$ satisfies (S), (CB) with $\beta^* = 1$, (VE) with $\beta_* = 1/8$, and $\widetilde{L}$ is bounded. By Lax–Milgram the variational problem of 4(a) has exactly one solution $u \in V_g$.

Mechanics reading. (VE) is "no rigid-body mode": the rim is held, so a membrane cannot move without stretching. (11) is the 2D version of Lecture 8's "pinned ends cannot move for free" — held on the left side, a deflection is controlled by its slope. $\beta^*/\beta_* = 8$ is the constant that Céa's lemma will carry into the error estimate of part (d).

Problem 4(c)

Problem. Develop a finite element method for this problem based on square elements. On each element the solution on the four corners is interpolated with $s(x, y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy$. Write out the basis functions and explicitly write out the linear system (uniform grid, same spacing in x and y).

Step 1 — grid, element, local coordinates

- $N \times N$ squares, $h = 2/N$, nodes $x_i = -1 + ih$, $y_j = -1 + jh$, $i, j = 0, \ldots, N$. Interior nodes $1 \le i, j \le N - 1$: $(N-1)^2$ unknowns. Boundary nodes carry the known value $g(x_i, y_j)$.
- Element $K_{ij} = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$, local coordinates $\xi = (x - x_i)/h$, $\eta = (y - y_j)/h \in [0, 1]$, corners numbered counter-clockwise from (x_i, y_j) (Figure 1).

Step 2 — the four corner values fix s; the shape functions

- Put $x = x_i + h\xi$, $y = y_j + h\eta$ into s and collect powers:

$$s = \underbrace{\alpha_1 + \alpha_2 x_i + \alpha_3 y_j + \alpha_4 x_i y_j}_{c_1} + \underbrace{h(\alpha_2 + \alpha_4 y_j)}_{c_2}\,\xi + \underbrace{h(\alpha_3 + \alpha_4 x_i)}_{c_3}\,\eta + \underbrace{h^2 \alpha_4}_{c_4}\,\xi\eta. \tag{15}$$

  Read bottom-up, $\alpha_4, \alpha_3, \alpha_2, \alpha_1$ follow from c_4, c_3, c_2, c_1 one at a time, so the α's and the c's carry the same information.

- Evaluate (15) at the four corners, $s_1 = s(0,0)$, $s_2 = s(1,0)$, $s_3 = s(1,1)$, $s_4 = s(0,1)$: $s_1 = c_1$, $s_2 = c_1 + c_2$, $s_4 = c_1 + c_3$, $s_3 = c_1 + c_2 + c_3 + c_4$. Solved one at a time,

$$c_1 = s_1, \quad c_2 = s_2 - s_1, \quad c_3 = s_4 - s_1, \quad c_4 = s_3 - s_2 - s_4 + s_1, \tag{16}$$

  a unique interpolant for any four corner values.

- Put (16) back into (15) and collect by corner value: $s = \sum_{a=1}^{4} s_a N_a(\xi, \eta)$ with

$$\boxed{N_1 = (1-\xi)(1-\eta), \qquad N_2 = \xi(1-\eta), \qquad N_3 = \xi\eta, \qquad N_4 = (1-\xi)\eta} \tag{17}$$

  Each is 1 at its own corner and 0 at the other three.

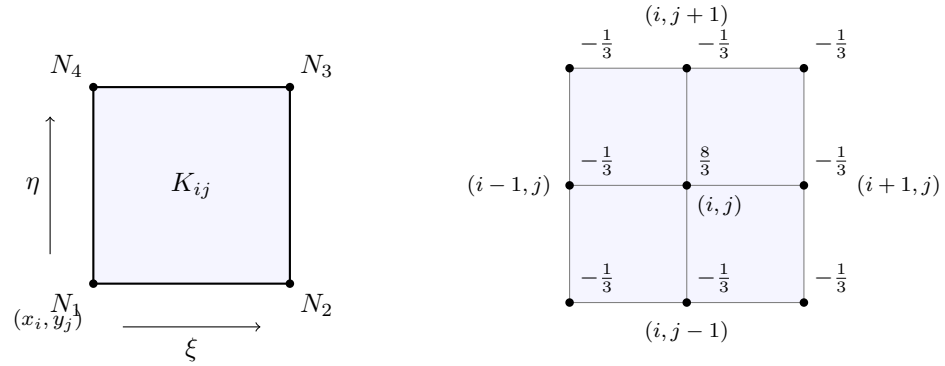


Figure 1: Left: one element and its four shape functions (17). Right: the four elements around an interior node (i, j) — the support of φ_{ij} — with the coefficients of its row of the stiffness matrix, (24).

Step 3 — continuity, and the global basis [Lecture 4: 1D hats; HW3 Problem 3: a jump puts a delta in the derivative]

- On the right edge of K_{ij}, $\xi = 1$ and (17) gives $s = s_2(1-\eta) + s_3\eta$: only the two corner values on that edge enter. The neighbour $K_{i+1,j}$ produces the same line from the same two values on its left edge. So s is continuous across every edge, ∇s has no delta, $\iint_\Omega |\nabla s|^2 < \infty$: the method is **conforming**, $V_h \subset$ the spaces of (6).
- Glue the N_a of the four squares around node (i,j) (Figure 1, right). On K_{ij} the node is corner 1, value $(1-\xi)(1-\eta)$; on $K_{i-1,j}$ it is corner 2, value $\xi(1-\eta)$; and so on. In every case the x-factor is Lecture 4's hat φ_i on the x-grid and the y-factor is φ_j on the y-grid:

$$\boxed{\varphi_{ij}(x,y) = \varphi_i(x)\,\varphi_j(y), \qquad \varphi_i(x) = \begin{cases} 1 - |x - x_i|/h, & |x - x_i| \le h, \\ 0, & \text{otherwise.} \end{cases}} \tag{18}$$

- Trial and test functions, with the boundary values as Lecture 5's lift:

$$U = \underbrace{\sum_{(k,l)\in\partial\Omega} g(x_k, y_l)\,\varphi_{kl}}_{\text{known, carries } u=g} + \sum_{i,j=1}^{N-1} \xi_{ij}\,\varphi_{ij}, \qquad v = \varphi_{ij}, \quad 1 \le i,j \le N-1. \tag{19}$$

(FEM) Find U of the form (19) such that $\displaystyle\iint_\Omega \nabla U \cdot \nabla\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y$ for every interior node (i,j).

Step 4 — the element stiffness matrix

- $\partial/\partial x = (1/h)\,\partial/\partial\xi$, $\partial/\partial y = (1/h)\,\partial/\partial\eta$ and $\mathrm{d}x\,\mathrm{d}y = h^2\,\mathrm{d}\xi\,\mathrm{d}\eta$, so

$$k^e_{ab} = \iint_K \nabla N_a \cdot \nabla N_b\,\mathrm{d}x\,\mathrm{d}y = \underbrace{\tfrac{1}{h^2}\cdot h^2}_{=1} \int_0^1\!\!\int_0^1 \big(N_{a,\xi}N_{b,\xi} + N_{a,\eta}N_{b,\eta}\big)\,\mathrm{d}\xi\,\mathrm{d}\eta. \tag{20}$$

  h cancels: in 2D the stiffness of a square does not depend on its size.
- Every N_a in (17) is a product of $1-\xi$ or ξ with $1-\eta$ or η, so (20) splits into 1D integrals:

$$\int_0^1 (1-\xi)^2\mathrm{d}\xi = \tfrac{1}{3}, \quad \int_0^1 \xi(1-\xi)\,\mathrm{d}\xi = \tfrac{1}{2} - \tfrac{1}{3} = \tfrac{1}{6}, \quad \big[(1-\xi)'\big]^2 = 1, \quad (1-\xi)'\,\xi' = -1. \tag{21}$$

- Three kinds of pair, each "x-part $\times$ y-part + y-part $\times$ x-part":

$$\underbrace{1\cdot\tfrac{1}{3} + \tfrac{1}{3}\cdot 1 = \tfrac{4}{6}}_{a=b}, \qquad \underbrace{(-1)\cdot\tfrac{1}{3} + \tfrac{1}{6}\cdot 1 = -\tfrac{1}{6}}_{a,b \text{ share an edge}}, \qquad \underbrace{(-1)\cdot\tfrac{1}{6} + \tfrac{1}{6}\cdot(-1) = -\tfrac{2}{6}}_{a,b \text{ opposite}}, \tag{22}$$

$$\boxed{\boldsymbol{k}^e = \frac{1}{6}\begin{bmatrix} 4 & -1 & -2 & -1 \\ -1 & 4 & -1 & -2 \\ -2 & -1 & 4 & -1 \\ -1 & -2 & -1 & 4 \end{bmatrix}} \tag{23}$$

  Each row sums to 0: a constant has no gradient, hence no energy.

Step 5 — the linear system [Lecture 13, eq. (1.236): $A_{ij} = B(\varphi_i, \varphi_j)$; Lecture 5, eq. (1.66): known values to the RHS; Lecture 4: trapezoidal load]

- **Row of an interior node** (i, j)**.** It sits in 4 squares as a corner, shares each edge-neighbour with 2 squares and each diagonal neighbour with 1. From (22): $4 \cdot \frac{4}{6} = \frac{8}{3}$, $2 \cdot (-\frac{1}{6}) = -\frac{1}{3}$, $1 \cdot (-\frac{2}{6}) = -\frac{1}{3}$:

$$\boxed{\frac{1}{3}\Big[\,8\,U_{i,j} - \sum_{(k,l)\in\mathcal{N}_{ij}} U_{k,l}\Big] = b_{ij}, \qquad 1 \le i, j \le N-1,} \tag{24}$$

  $\mathcal{N}_{ij}$ the eight neighbours $k \in \{i-1, i, i+1\}$, $l \in \{j-1, j, j+1\}$, $(k,l) \neq (i,j)$, and $U_{k,l} = g(x_k, y_l)$ whenever (k,l) is a boundary node.

- **Load.** $b_{ij} = \iint_\Omega f\varphi_{ij}$. The trapezoidal rule of Lecture 4 on each square, $\iint_K F \approx \frac{h^2}{4}\sum_{\text{corners}} F$, sees $f\varphi_{ij}$ only at corner (i, j), where $\varphi_{ij} = 1$; four squares:

$$b_{ij} \approx 4 \cdot \tfrac{h^2}{4}\, f(x_i, y_j) = h^2 f_{ij}. \tag{25}$$

- **Matrix form.** Order the unknowns row by row, $\boldsymbol{U} = (U_{1,1}, \dots, U_{N-1,1}, U_{1,2}, \dots, U_{N-1,N-1})^T$. Move every boundary value in (24) to the right:

$$\boxed{\begin{gathered}\boldsymbol{A}\,\boldsymbol{U} = \boldsymbol{b} + \boldsymbol{r}, \qquad \boldsymbol{A} = \frac{1}{3}\begin{bmatrix} \boldsymbol{T} & \boldsymbol{S} & & \\ \boldsymbol{S} & \boldsymbol{T} & \ddots & \\ & \ddots & \ddots & \boldsymbol{S} \\ & & \boldsymbol{S} & \boldsymbol{T} \end{bmatrix}, \\ \boldsymbol{T} = \begin{bmatrix} 8 & -1 & & \\ -1 & 8 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 8 \end{bmatrix}, \qquad \boldsymbol{S} = \begin{bmatrix} -1 & -1 & & \\ -1 & -1 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & -1 \end{bmatrix}\end{gathered}} \tag{26}$$

  $\boldsymbol{A}$ is $(N-1)^2 \times (N-1)^2$, made of $N-1$ block rows; $\boldsymbol{T}, \boldsymbol{S}$ are $(N-1) \times (N-1)$; $\boldsymbol{T}$ couples a node to its own grid row, $\boldsymbol{S}$ to the rows above and below.

- $r_{ij} = \frac{1}{3}\sum g(x_k, y_l)$ over the boundary nodes among the eight neighbours of (i, j). For $g = 1$ (part (d)):

$$r_{ij} = \begin{cases} 5/3, & (i,j) \text{ next to a corner (5 boundary neighbours)}, \\ 1, & (i,j) \text{ next to one side (3 boundary neighbours)}, \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

- $\boldsymbol{A}$ is symmetric positive definite by Lecture 13's claim: (S) and (VE) of 4(b) hold, whatever the element shape.

Same as finite differences? Divide (24) by h^2 and use (25): $\big[8U_{i,j} - \sum_{\mathcal{N}_{ij}} U_{k,l}\big]/(3h^2) = f_{ij}$, a nine-point difference formula for $-\Delta u$. This is Lecture 4's "CG(1) with trapezoidal load is the central difference method", one dimension up: the 2D element couples diagonal neighbours too, which the five-point formula does not.